

Jackson Vaughn

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RESEARCH EXPERIENCE

Master's Thesis, Clean Lab, Bristol

09.09.2024 - Present

Supervisors: Prof. Dr. Laura Robinson, Prof. Dr. Erica Hendy, Dr. Joseph Stewart

Working Title: "Controls Over the Geochemistry of Galapagos Stylasterids"

Sampled and analyzed over 60 stylasterid (deep-sea) corals, varying in species, taken from the Galapagos using ICP-MS analysis. Analyzed the trace element composition and mineralogy of the samples and compared this with their respective hydrographic parameters stored in large datasets. Assessed the potential use of Stylasterids as paleo-oceanographic proxies. Investigated the resistance of Stylasterid corals to ongoing ocean acidification. Integrated bathymetric and remote-sensing data with trace-element data to create a website for interactive data visualization during the Goldschmidt 2025 conference. Research is expected to contribute to publications in the near future.

Sensor Development, Workshop, Bristol

01.01.2025 - Present

Assisting a collegial PhD student in her research on peatlands by designing and building a low-cost CO₂ flux sensor for field deployment.

Lab Assistant, BOSCORF, NOC, Southampton

11.11.2024 - 13.11.2024

Assisted in Galapagos sediment core analysis (XRF, X-ray tomography, MSCL-XYZ) and archival.

Bachelor's Thesis, Ocean Lab, Bremen

03.11.2022 - 31.08.2023

Supervisor: Prof. Dr. Michael Bau

Title: "Rare Earth Element and Yttrium Partitioning in Modern Microbialites from Lake Eyasi, Tanzania"

Collected major and trace element data of Holocene-aged microbialites (biogenic carbonate), and analyzed aforementioned data through the use of Python scripts. Compared sample data to that of their respective ambient water. Suggested hypotheses regarding their relationships while assessing the potential to use microbialites as a geochemical proxy for the ambient water in which they were formed.

Research Cruise North Sea, FS Heincke, North Sea & Helgoland

25.04.2022 - 27.04.2022

Lead Instructors: Dr. Gernot Nehrke & Prof. Dr. Vikram Unnithan

Designed and led a small sediment research project: sediment sampling (van veen grab, box core, MUC), sediment classification, and statistical analysis of sediment grain size/shape. Additionally; became acquainted with instruments including parametric and multibeam echosounders.

Assistant Scientist, AWI, Bremerhaven

08.02.2022 - 01.10.2022

Supervisor: Prof. Dr. Jelle Bijma

Assisted in preparing enhanced weathering experiments involving the application of basalt to agricultural land and performed subsequent field work including taking *in situ* measurements of soil pore water pH and conductivity. Assisted in sample preparation for laboratory analysis of dissolved inorganic carbon and organic alkalinity of aqueous samples. Assisted in the maintenance of an experiment to investigate olivine dissolution under shear stress in aqueous environments. Wrote a research proposal for a greenhouse experiment to test the rate of dissolution of basalt powder in soil with and without organic matter, and supervised said experiment.

SKILLS

- ❖ Python
- ❖ ICP-MS analysis
- ❖ LaTeX

EDUCATION

University of Bristol — Bristol, United Kingdom

Master of Science by Research: Geology

Sep 2024 - Present, Expected Graduation May 2026

Jacobs University Bremen — Bremen, Germany

Bachelor of Science: Earth and Environmental Science

September 2020 - June 2024

Decatur High School —

Decatur, GA, USA

August 2016 - May 2020

LANGUAGES

English: Native

German: Advanced (C1)

PROFILE

Website

<https://jackson-vaughn.netlify.app/map>

GitHub

<https://github.com/Jacksonrv>

REFERENCES

Dr. Joseph Stewart
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Dr. Erica Hendy
e.hendy@bristol.ac.uk